

Debugging Report: Netlist Fixes for alcazar.v

1. Issues Identified in the Original Netlist

Broken Signal Names:

- Some signal names were split across lines (e.g., n861 and 34 instead of n86134).

Fixed using:

```
...  
perl -0777 -i -pe 's/\n(?:?!(:module|endmodule)\b)[^\t]//g' ../inputs/RTL/alcazar_fixed.v  
...
```

Incorrect Pin Mappings in Standard Cells:

- **NAND/NOR Cells:** Had .A(instead of .AN(for inverted inputs.
- **MUX Cells:** Had .sel(instead of .S0(for select pins.
- Fixed using sed:

```
...  
sed -i '/NAND[0-9]\+BX[0-9]\+/s/\.A(/\.AN(/g' ../inputs/RTL/alcazar_fixed.v  
sed -i '/NOR[0-9]\+BX[0-9]\+/s/\.A(/\.AN(/g' ../inputs/RTL/alcazar_fixed.v  
sed -i '/MX[0-9]\+X[0-9]\+/s/\.sel(/\.S0(/g' ../inputs/RTL/alcazar_fixed.v  
sed -i '/MXI[0-9]\+X[0-9]\+/s/\.sel(/\.S0(/g' ../inputs/RTL/alcazar_fixed.v  
...
```

2. Tempus Analysis Setup

Clock Definition:

- Clock period = **5 ns** (200 MHz).
- Defined in alcazar_constraints.sdc:

```
...  
create_clock -name clk -period 5 [get_ports clk]  
...
```

- **Library Used:**
 - Cadence RAK standard cell library (slow.lib).

3. Results After Fixes

- **Original Errors:**
 - Broken signal names → Syntax errors in Tempus.
 - Incorrect pin mappings → Cell instantiation errors.
- **After Fixes:**

- **No syntax errors** in `alcazar_fixed.v`.
- **Correct cell mappings** (NAND/NOR/MUX pins fixed).
- Tempus analysis runs **without errors**.

Files:

Fixed Netlist: `alcazar_fixed.v`

SDC File: `alcazar_constraints.sdc`